Shaheed Zulfikar Ali Bhutto Institute of Science & Technology University

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 7.5

Obtained Marks: _____

Artificial Neural Networks Lab

Project: AI-Based Urdu Speech-to-English Translation System

Submitted To: **Hassan Mujtaba**

Student Name: **Mehma Razaq**

Registration Number: **24108214**

1. Introduction

Audio recordings are widely used in education, communication, and everyday life. However, understanding Urdu audio can be difficult for users who need its content in written or English form.

Project ECHO addresses this problem by automating Urdu speech transcription and English translation. It combines pre-trained AI models with a simple web interface to make the process convenient and accessible.

2. Problem Statement

Manually transcribing Urdu recordings and translating them into English can be time-consuming. The project aims to develop an automated system that performs both tasks and displays the results through a user-friendly web application.

3. Objectives

- Convert Urdu audio into written Urdu text.
- Translate Urdu transcription into English.
- Integrate pre-trained AI models into one application.
- Develop a user-friendly interface using Flask.
- Reduce the effort required for manual transcription and translation.

4. Technologies Used

Technology	Purpose
Python	Main programming language
Whisper Urdu	Urdu speech recognition
NLLB-200	Urdu-to-English translation
PyTorch	Model execution
Hugging Face Transformers	Loading pre-trained models
Librosa	Audio preprocessing
Flask	Web application
HTML and CSS	Frontend design
Google Colab	Running AI models
Gradio Client	Connecting Flask to the model service

5. Methodology

The system processes audio through the following stages:

1. **Audio Upload:** The user uploads an Urdu audio file through the Flask interface.
2. **Audio Preprocessing:** Librosa loads the recording, converts it to mono, and resamples it to 16 kHz.
3. **Speech Recognition:** The Whisper-based model converts the audio into Urdu text.
4. **Machine Translation:** The NLLB model translates the Urdu transcription into English.
5. **Result Display:** The Flask application displays both outputs separately.

Long recordings are divided into smaller segments to make processing more manageable.

6. System Implementation

Two pre-trained models are used in the project:

- **Speech Recognition Model:** `kingabzpro/whisper-large-v3-urdu`
- **Translation Model:** `facebook/nllb-200-distilled-1.3B`

The models run in Google Colab. Python functions handle transcription, translation, and result generation. Gradio exposes the processing function through a temporary endpoint, and the Gradio Client connects this service to the local Flask application.

The frontend allows users to upload supported audio files, start processing, and view the generated Urdu and English text.

7. Results and Discussion

During development, the transcription and translation pipeline was successfully executed on the tested audio. The system generated Urdu transcription and its corresponding English translation.

The results demonstrate that the selected models can be integrated into a single application. However, transcription and translation quality may vary depending on pronunciation, background noise, recording quality, and audio content.

A formal accuracy percentage has not been reported because quantitative evaluation has not yet been performed.

8. Limitations

- Background noise and unclear speech may affect transcription.
- Machine-generated translations may require manual correction.
- Long recordings may take additional processing time.
- The system depends on an active Google Colab runtime and a temporary Gradio link.
- The application has not been evaluated using a large benchmark dataset.

9. Future Enhancements

Future improvements may include permanent backend hosting, downloadable transcription and translation files, faster processing, improved error handling, formal accuracy evaluation, and support for additional languages.

10. Conclusion

Project ECHO demonstrates an AI-based approach to Urdu speech recognition and English translation. By combining Whisper, NLLB, and a Flask web interface, the system provides a practical way to convert Urdu audio into written Urdu and English text. Further testing and permanent hosting could improve its reliability and usability.



